

## FISTER SIGMUNDSTAD, Norway

WMO: 014220

Lat: 59.1597N

Lon: 6.0364E

Elev: 25

StdP: 101.03

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.0       | -5.3 | -14.8                         | 1.0 | -4.3 | -12.7 | 1.3 | -3.8 | 11.5                  | 5.0  | 9.7 | 5.1  | 2.9                   | 100  | 0.554 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.2                    | 24.6            | 16.9 | 22.4 | 16.1 | 20.6 | 15.3 | 18.0                | 22.3 | 17.0 | 21.0 | 16.1 | 19.5 | 2.2                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.2                            | 11.5 | 19.7 | 15.2 | 10.8 | 18.8 | 14.4 | 10.3 | 18.0 | 50.9          | 22.1 | 47.9 | 21.0 | 45.0 | 19.5 | 21.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 8.5           | 7.3  | 6.3 | DB                         | -6.8 | 28.3               | 2.4 | 2.5                                                | -8.5 | 30.1       | -9.9  | 31.6       | -11.2 | 33.0       | -12.9 | 34.8 | (4) |
| (5)               |      |     | WB                         | -8.3 | 19.0               | 2.2 | 1.4                                                | -9.9 | 20.0       | -11.2 | 20.9       | -12.4 | 21.7       | -14.1 | 22.7 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.7         | 2.0   | 2.4   | 4.2   | 7.3   | 10.9  | 13.4  | 16.1  | 15.6  | 13.2  | 9.2   | 5.7   | 3.5   | (6)  |  |
|                                                                        | DBStd       | 5.80        | 3.48  | 3.29  | 2.60  | 2.96  | 3.62  | 2.54  | 2.86  | 2.19  | 2.38  | 2.76  | 3.50  | 3.73  | (7)  |  |
|                                                                        | HDD10.0     | 1148        | 248   | 214   | 179   | 91    | 29    | 2     | 0     | 0     | 3     | 47    | 133   | 201   | (8)  |  |
|                                                                        | HDD18.3     | 3556        | 507   | 447   | 438   | 329   | 233   | 149   | 84    | 92    | 154   | 284   | 380   | 459   | (9)  |  |
|                                                                        | CDD10.0     | 659         | 0     | 0     | 0     | 11    | 58    | 105   | 189   | 172   | 99    | 22    | 3     | 1     | (10) |  |
|                                                                        | CDD18.3     | 26          | 0     | 0     | 0     | 0     | 3     | 3     | 14    | 5     | 1     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 123         | 0     | 0     | 0     | 0     | 11    | 14    | 80    | 17    | 2     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 19          | 0     | 0     | 0     | 0     | 1     | 1     | 15    | 2     | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.7         | 3.2   | 2.8   | 2.8   | 2.6   | 2.6   | 2.6   | 2.2   | 2.4   | 2.5   | 2.8   | 2.9   | 3.3   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1764        | 186   | 154   | 138   | 93    | 81    | 95    | 116   | 147   | 176   | 203   | 193   | 198   | (15) |  |
|                                                                        | PrecMax     | 2320        | 376   | 376   | 306   | 158   | 203   | 165   | 260   | 268   | 320   | 407   | 373   | 371   | (16) |  |
|                                                                        | PrecMin     | 1205        | 0     | 0     | 6     | 40    | 4     | 19    | 50    | 36    | 54    | 53    | 22    | 18    | (17) |  |
|                                                                        | PrecStd     | 288         | 101   | 98    | 71    | 33    | 46    | 46    | 51    | 54    | 79    | 82    | 102   | 110   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 9.0   | 9.9   | 11.8  | 18.5  | 24.5  | 25.4  | 28.7  | 25.7  | 22.4  | 18.4  | 14.7  | 11.2  | (19) |  |
|                                                                        |             | MCWB        | 7.1   | 6.7   | 7.9   | 10.3  | 15.5  | 15.5  | 18.2  | 17.5  | 15.7  | 13.7  | 10.9  | 8.9   | (20) |  |
|                                                                        | 2%          | DB          | 8.2   | 8.3   | 9.7   | 16.0  | 21.8  | 21.9  | 25.6  | 22.6  | 19.4  | 15.9  | 12.4  | 10.0  | (21) |  |
|                                                                        |             | MCWB        | 6.7   | 6.4   | 6.8   | 9.6   | 14.2  | 15.6  | 17.6  | 15.9  | 14.4  | 12.5  | 9.3   | 8.6   | (22) |  |
|                                                                        | 5%          | DB          | 7.4   | 7.5   | 8.3   | 14.3  | 19.1  | 19.4  | 23.0  | 20.7  | 18.3  | 14.1  | 11.1  | 9.1   | (23) |  |
|                                                                        |             | MCWB        | 6.1   | 5.9   | 5.9   | 9.0   | 13.2  | 14.4  | 17.2  | 15.4  | 13.9  | 11.0  | 8.7   | 7.6   | (24) |  |
|                                                                        | 10%         | DB          | 6.5   | 6.7   | 7.6   | 12.1  | 16.8  | 17.6  | 20.8  | 19.3  | 17.1  | 12.9  | 10.2  | 8.2   | (25) |  |
|                                                                        |             | MCWB        | 5.2   | 5.2   | 5.4   | 8.1   | 11.6  | 13.4  | 16.3  | 14.9  | 13.3  | 10.2  | 7.9   | 6.4   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 7.9   | 7.7   | 8.7   | 11.2  | 16.6  | 17.4  | 20.0  | 18.7  | 17.2  | 14.9  | 11.4  | 9.7   | (27) |  |
|                                                                        |             | MCDB        | 8.5   | 8.6   | 10.1  | 15.9  | 23.0  | 22.4  | 26.0  | 23.4  | 21.1  | 17.0  | 13.5  | 10.4  | (28) |  |
|                                                                        | 2%          | WB          | 7.1   | 7.0   | 7.4   | 10.2  | 14.9  | 15.9  | 18.6  | 17.3  | 15.7  | 13.0  | 10.2  | 8.8   | (29) |  |
|                                                                        |             | MCDB        | 7.8   | 8.0   | 8.7   | 14.5  | 20.2  | 20.1  | 23.7  | 20.8  | 18.2  | 15.1  | 11.9  | 9.7   | (30) |  |
|                                                                        | 5%          | WB          | 6.3   | 6.3   | 6.7   | 9.5   | 13.5  | 14.8  | 17.6  | 16.2  | 14.6  | 12.0  | 9.4   | 7.8   | (31) |  |
|                                                                        |             | MCDB        | 7.2   | 7.3   | 8.0   | 13.3  | 18.0  | 18.4  | 21.7  | 19.1  | 17.4  | 13.7  | 10.9  | 8.9   | (32) |  |
|                                                                        | 10%         | WB          | 5.3   | 5.5   | 6.0   | 8.6   | 12.3  | 13.9  | 16.6  | 15.4  | 13.7  | 10.7  | 8.2   | 6.7   | (33) |  |
|                                                                        |             | MCDB        | 6.4   | 6.6   | 7.2   | 11.7  | 16.1  | 17.1  | 20.5  | 18.3  | 16.2  | 12.3  | 9.8   | 7.9   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 3.7   | 4.0   | 4.7   | 6.2   | 6.7   | 6.4   | 6.2   | 6.0   | 5.5   | 4.9   | 4.0   | 3.8   | (35) |  |
|                                                                        |             | MCDBR       | 4.4   | 4.1   | 5.1   | 9.2   | 10.4  | 9.5   | 10.0  | 8.2   | 7.2   | 6.0   | 4.4   | 4.1   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 4.0   | 3.6   | 3.7   | 4.9   | 5.3   | 5.0   | 4.6   | 4.1   | 4.0   | 4.0   | 3.5   | 4.0   | (37) |  |
|                                                                        |             | MCWBR       | 4.1   | 4.2   | 4.7   | 7.8   | 9.6   | 8.1   | 8.7   | 6.8   | 6.4   | 4.9   | 4.3   | 4.2   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.280 | 0.309 | 0.335 | 0.369 | 0.364 | 0.358 | 0.373 | 0.370 | 0.349 | 0.329 | 0.292 | 0.259 | (40) |  |
|                                                                        |             | taud        | 2.166 | 2.259 | 2.281 | 2.271 | 2.398 | 2.457 | 2.434 | 2.437 | 2.492 | 2.448 | 2.271 | 2.104 | (41) |  |
|                                                                        | Edn at Noon | Edn at Noon | 515   | 702   | 798   | 831   | 866   | 877   | 853   | 826   | 787   | 685   | 506   | 396   | (42) |  |
|                                                                        |             | Edh at Noon | 46    | 71    | 93    | 111   | 106   | 101   | 101   | 94    | 76    | 59    | 42    | 33    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 0.31  | 0.91  | 2.17  | 3.87  | 4.96  | 5.38  | 4.77  | 3.69  | 2.36  | 1.15  | 0.42  | 0.20  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.06  | 0.18  | 0.34  | 0.52  | 0.67  | 0.62  | 0.54  | 0.32  | 0.23  | 0.17  | 0.07  | 0.03  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (37 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -116    | N/A     | N/A     |

Nomenclature: See separate page